Vishhvaan Gopalakrishnan

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EDUCATION

CLEVELAND CLINIC CASE WESTERN RESERVE U.
Medical Doctor (M.D.)
w/ Research Qual.
May 2020 | Cleveland, OH

UNIVERSITY OF MICHIGAN AT ANN ARBOR
M.S.E. and B.S.E. in Biomedical Engineering
May 2013 | Ann Arbor, MI
Summa Cum Laude

MEDICAL TRAINING

ROTATIONS
Internal Medicine (AI), Pediatrics (AI), Surgery, Emergency Medicine, Neurology, OB/GYN, Psychiatry, Family Medicine
(AI is Acting Intern denoting more responsibility)

SKILLS

PROGRAMMING
Over 5000 lines:
Python • Shell • Docker
R • Matlab • CSS • LaTeX
Familiar:
C • C++ • Java • MySQL

MEDICAL
EMR Usage (EPIC and Cerna)
Medical Equipment Usage and Need
Hospital Work Flow

ACTIVITIES

- CWRU Free Clinic (Business Leader)
- Med School Tech Committee (Member)
- M-HEAL (Eng. Outreach) (Member)
- U. Mich Engineering Council (Leadership)

LINKS

Github:// **vishhvaan**
LinkedIn:// **vishhvaan**
Twitter:// **@vishhvaan**

RESEARCH

THEORY DIVISION - LAB STUDYING EVOLUTION | Researcher
Dec 2018 - Apr 2020 | Cleveland, OH
Worked with **Jacob Scott, M.D.** and other lab members to create the **EVE**, an open source bioreactor to study evolutionary paths of microrganisms and cancer. I desinged 3D parts, all the circuits and PCB, and a software suite in Python that enables users to design and monitor experiments done in the machine.

HEART AND VASCULAR INSTITUTE GENOMICS | Researcher
Jul 2015 – Jul 2017 | Cleveland, OH
Investigated genomic etiologies for the difference between the incidence of atrial fibriation between men and women with **Mina Chung M.D.**. Project was heavy in statistics (analysis with the R language) and incorporated basic machine learning in TensorFlow.

THEREPUTIC ULTRASOUND GROUP | Thesis Project
Jan 2012 – Apr 2013 | Ann Arbor, MI
Designed and prototyped Acoustic Coupler for a novel clinical procedure called histotripsy with my team in the **Xu lab**. Work led to designs that were used for patent applications by the lab.

AWARDS

2019	1/15 Nationally	Hematology Opportunities for the Next Generation of Research Scientists (HONORS) by The American Society of Hematology
2013	32 Med. Students	Cleveland Clinic Medical School Full Tuition Scholarship
2012	U. Mich.	University of Michigan Honors College of Engineering Dean's List

EXPERIENCE

LIVESCRIBE INC. | Research and Development Engineering Intern
June 2010 – Sep 2010 | Oakland, CA
- Worked on development of cloud-based sync software for the Livescribe smartpen.
- Worked on database management and introduced to server-client relationships.

SIEMENS INC. | Research and Development Engineering Intern
June 2009 – Sep 2009 | Buffalo Grove, IL
- Worked with a team to develop a controller for a rainwater catchment device.
- Hardware engineering focused project which incuded circuit and printed circuit board (PCB) design.

PUBLICATIONS

[1] V. Gopalakrishnan, N. P. Krishnan, E. McClure, J. Pelesko, D. Crozier, D. F. Williamson, N. Webster, D. Ecker, D. Nichol, S. Mandal, R. A. Bonomo, and J. G. Scott. A Low-Cost, Open Source, Self-Contained Bacterial EVolutionary biorEactor (EVE). *bioRxiv*, 2019.